

PakGenie Packaging Knowledge Reference

Dataset-grounded reference for food commodity traits, packaging materials, storage conditions, and observed shelf life.

Source scope

This reference uses only the supplied `packaging_classifier_dataset.csv` and `shelf_life_regressor_dataset.csv`. Values and recommendations are reported as observed in those files. Where a packaging material has more than one observed OTR or WVTR value, the full observed range is reported rather than collapsing it to an invented single value.

1. Introduction

PakGenie is represented here as a data reference for connecting food traits with packaging recommendations and observed shelf-life records. The classifier dataset contains 25 unique food types and their recorded respiration, ethylene, moisture, chilling, and mechanical-fragility traits, plus recommended packaging. The shelf-life dataset adds pH, storage temperature, relative humidity, packaging barrier values, and observed shelf life in days.

Glossary

OTR: Oxygen Transmission Rate, recorded in $\text{cc/m}^2/\text{day}$. In this reference, lower OTR is described as a better oxygen barrier.

WVTR: Water Vapor Transmission Rate, recorded in $\text{g/m}^2/\text{day}$. In this reference, lower WVTR is described as less water-vapor transmission.

Respiration rate: The dataset's categorical indication of how active respiration is for a commodity: Very Low, Low, Medium, High, or Very High.

Ethylene sensitivity: The dataset's categorical sensitivity field: Low, Medium, or High.

Chilling sensitivity: Whether the dataset marks a commodity as sensitive to chilling, recorded as Yes or No.

Mechanical fragility: The dataset's categorical indication of physical fragility: Low, Medium, or High.

MAP: Modified Atmosphere Packaging, appearing as the packaging material label 'MAP High Barrier' in the supplied data.

2.1 Strawberry

- Respiration rate: **Very High** — The very high respiration rate indicates a strongly active commodity in this dataset, so its packaging recommendation is paired with a trait profile that requires the listed packaging choice.
- Ethylene production / sensitivity: **Low / Low** — Ethylene production is Low and ethylene sensitivity is Low in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **91%**
- pH level: **3.5**
- Chilling sensitivity: **No**
- Mechanical fragility: **High**
- Recommended packaging: **PP Perforated Clamshell**

Summary: Strawberry is recorded with a very high respiration rate, low ethylene production, low ethylene sensitivity, and 91% moisture content. Its chilling sensitivity is no and its mechanical fragility is high. In the classifier records, this trait combination is paired with PP Perforated Clamshell. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.2 Blueberry

- Respiration rate: **High** — The high respiration rate indicates active respiration in this dataset, and the recommended package is the one paired with this trait profile.
- Ethylene production / sensitivity: **Low / Low** — Ethylene production is Low and ethylene sensitivity is Low in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **84%**
- pH level: **3.3**
- Chilling sensitivity: **No**
- Mechanical fragility: **High**
- Recommended packaging: **LDPE Vented**

Summary: Blueberry is recorded with a high respiration rate, low ethylene production, low ethylene sensitivity, and 84% moisture content. Its chilling sensitivity is no and its mechanical fragility is high. In the classifier records, this trait combination is paired with LDPE Vented. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.3 Spinach

- Respiration rate: **Very High** — The very high respiration rate indicates a strongly active commodity in this dataset, so its packaging recommendation is paired with a trait profile that requires the listed packaging choice.
- Ethylene production / sensitivity: **Very Low / High** — Ethylene production is Very Low and ethylene sensitivity is High in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **91.5%**
- pH level: **6.5**
- Chilling sensitivity: **No**
- Mechanical fragility: **Medium**
- Recommended packaging: **LDPE Perforated**

Summary: Spinach is recorded with a very high respiration rate, very low ethylene production, high ethylene sensitivity, and 91.5% moisture content. Its chilling sensitivity is no and its mechanical fragility is medium. In the classifier records, this trait combination is paired with LDPE Perforated. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.4 Tomato

- Respiration rate: **Medium** — The medium respiration rate indicates an intermediate respiration level in this dataset, alongside the listed packaging recommendation.
- Ethylene production / sensitivity: **High / Medium** — Ethylene production is High and ethylene sensitivity is Medium in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **94%**
- pH level: **4.3**
- Chilling sensitivity: **Yes**
- Mechanical fragility: **Medium**
- Recommended packaging: **LDPE**

Summary: Tomato is recorded with a medium respiration rate, high ethylene production, medium ethylene sensitivity, and 94% moisture content. Its chilling sensitivity is yes and its mechanical fragility is medium. In the classifier records, this trait combination is paired with LDPE. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.5 Broccoli

- Respiration rate: **Very High** — The very high respiration rate indicates a strongly active commodity in this dataset, so its packaging recommendation is paired with a trait profile that requires the listed packaging choice.
- Ethylene production / sensitivity: **Low / High** — Ethylene production is Low and ethylene sensitivity is High in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **90%**
- pH level: **6.3**
- Chilling sensitivity: **No**
- Mechanical fragility: **Medium**
- Recommended packaging: **LDPE Perforated**

Summary: Broccoli is recorded with a very high respiration rate, low ethylene production, high ethylene sensitivity, and 90% moisture content. Its chilling sensitivity is no and its mechanical fragility is medium. In the classifier records, this trait combination is paired with LDPE Perforated. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.6 Mango

- Respiration rate: **Medium** — The medium respiration rate indicates an intermediate respiration level in this dataset, alongside the listed packaging recommendation.
- Ethylene production / sensitivity: **High / High** — Ethylene production is High and ethylene sensitivity is High in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **82%**
- pH level: **4**
- Chilling sensitivity: **Yes**
- Mechanical fragility: **Medium**
- Recommended packaging: **Corrugated Board**

Summary: Mango is recorded with a medium respiration rate, high ethylene production, high ethylene sensitivity, and 82% moisture content. Its chilling sensitivity is yes and its mechanical fragility is medium. In the classifier records, this trait combination is paired with Corrugated Board. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.7 Banana

- Respiration rate: **High** — The high respiration rate indicates active respiration in this dataset, and the recommended package is the one paired with this trait profile.
- Ethylene production / sensitivity: **High / High** — Ethylene production is High and ethylene sensitivity is High in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **75%**
- pH level: **5**
- Chilling sensitivity: **Yes**
- Mechanical fragility: **Medium**
- Recommended packaging: **LDPE**

Summary: Banana is recorded with a high respiration rate, high ethylene production, high ethylene sensitivity, and 75% moisture content. Its chilling sensitivity is yes and its mechanical fragility is medium. In the classifier records, this trait combination is paired with LDPE. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.8 Apple

- Respiration rate: **Low** — The low respiration rate indicates less active respiration in this dataset, alongside the listed packaging recommendation.
- Ethylene production / sensitivity: **Medium / Medium** — Ethylene production is Medium and ethylene sensitivity is Medium in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **84%**
- pH level: **3.5**
- Chilling sensitivity: **No**
- Mechanical fragility: **Low**
- Recommended packaging: **LDPE**

Summary: Apple is recorded with a low respiration rate, medium ethylene production, medium ethylene sensitivity, and 84% moisture content. Its chilling sensitivity is no and its mechanical fragility is low. In the classifier records, this trait combination is paired with LDPE. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.9 Potato

- Respiration rate: **Low** — The low respiration rate indicates less active respiration in this dataset, alongside the listed packaging recommendation.
- Ethylene production / sensitivity: **Very Low / Low** — Ethylene production is Very Low and ethylene sensitivity is Low in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **79%**
- pH level: **5.8**
- Chilling sensitivity: **Yes**
- Mechanical fragility: **Low**
- Recommended packaging: **Corrugated Board**

Summary: Potato is recorded with a low respiration rate, very low ethylene production, low ethylene sensitivity, and 79% moisture content. Its chilling sensitivity is yes and its mechanical fragility is low. In the classifier records, this trait combination is paired with Corrugated Board. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.10 Carrot

- Respiration rate: **Low** — The low respiration rate indicates less active respiration in this dataset, alongside the listed packaging recommendation.
- Ethylene production / sensitivity: **Very Low / Medium** — Ethylene production is Very Low and ethylene sensitivity is Medium in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **88%**
- pH level: **6**
- Chilling sensitivity: **No**
- Mechanical fragility: **Low**
- Recommended packaging: **LDPE Perforated**

Summary: Carrot is recorded with a low respiration rate, very low ethylene production, medium ethylene sensitivity, and 88% moisture content. Its chilling sensitivity is no and its mechanical fragility is low. In the classifier records, this trait combination is paired with LDPE Perforated. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.11 Onion

- Respiration rate: **Very Low** — The very low respiration rate indicates limited respiration in this dataset, alongside the listed packaging recommendation.
- Ethylene production / sensitivity: **Very Low / Low** — Ethylene production is Very Low and ethylene sensitivity is Low in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **86%**
- pH level: **5.5**
- Chilling sensitivity: **No**
- Mechanical fragility: **Low**
- Recommended packaging: **Mesh Bag**

Summary: Onion is recorded with a very low respiration rate, very low ethylene production, low ethylene sensitivity, and 86% moisture content. Its chilling sensitivity is no and its mechanical fragility is low. In the classifier records, this trait combination is paired with Mesh Bag. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.12 Garlic

- Respiration rate: **Very Low** — The very low respiration rate indicates limited respiration in this dataset, alongside the listed packaging recommendation.
- Ethylene production / sensitivity: **Very Low / Low** — Ethylene production is Very Low and ethylene sensitivity is Low in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **60%**
- pH level: **6**
- Chilling sensitivity: **No**
- Mechanical fragility: **Low**
- Recommended packaging: **Mesh Bag**

Summary: Garlic is recorded with a very low respiration rate, very low ethylene production, low ethylene sensitivity, and 60% moisture content. Its chilling sensitivity is no and its mechanical fragility is low. In the classifier records, this trait combination is paired with Mesh Bag. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.13 Cabbage

- Respiration rate: **Low** — The low respiration rate indicates less active respiration in this dataset, alongside the listed packaging recommendation.
- Ethylene production / sensitivity: **Very Low / Medium** — Ethylene production is Very Low and ethylene sensitivity is Medium in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **92%**
- pH level: **5.8**
- Chilling sensitivity: **No**
- Mechanical fragility: **Low**
- Recommended packaging: **LDPE Perforated**

Summary: Cabbage is recorded with a low respiration rate, very low ethylene production, medium ethylene sensitivity, and 92% moisture content. Its chilling sensitivity is no and its mechanical fragility is low. In the classifier records, this trait combination is paired with LDPE Perforated. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.14 Lettuce

- Respiration rate: **High** — The high respiration rate indicates active respiration in this dataset, and the recommended package is the one paired with this trait profile.
- Ethylene production / sensitivity: **Very Low / High** — Ethylene production is Very Low and ethylene sensitivity is High in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **95%**
- pH level: **6**
- Chilling sensitivity: **No**
- Mechanical fragility: **Medium**
- Recommended packaging: **LDPE Perforated**

Summary: Lettuce is recorded with a high respiration rate, very low ethylene production, high ethylene sensitivity, and 95% moisture content. Its chilling sensitivity is no and its mechanical fragility is medium. In the classifier records, this trait combination is paired with LDPE Perforated. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.15 Cucumber

- Respiration rate: **Medium** — The medium respiration rate indicates an intermediate respiration level in this dataset, alongside the listed packaging recommendation.
- Ethylene production / sensitivity: **Low / High** — Ethylene production is Low and ethylene sensitivity is High in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **96%**
- pH level: **5.1**
- Chilling sensitivity: **Yes**
- Mechanical fragility: **Medium**
- Recommended packaging: **LDPE Shrink Wrap**

Summary: Cucumber is recorded with a medium respiration rate, low ethylene production, high ethylene sensitivity, and 96% moisture content. Its chilling sensitivity is yes and its mechanical fragility is medium. In the classifier records, this trait combination is paired with LDPE Shrink Wrap. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.16 Bell Pepper

- Respiration rate: **Low** — The low respiration rate indicates less active respiration in this dataset, alongside the listed packaging recommendation.
- Ethylene production / sensitivity: **Low / Low** — Ethylene production is Low and ethylene sensitivity is Low in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **92%**
- pH level: **5.2**
- Chilling sensitivity: **Yes**
- Mechanical fragility: **Medium**
- Recommended packaging: **LDPE**

Summary: Bell Pepper is recorded with a low respiration rate, low ethylene production, low ethylene sensitivity, and 92% moisture content. Its chilling sensitivity is yes and its mechanical fragility is medium. In the classifier records, this trait combination is paired with LDPE. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.17 Grapes

- Respiration rate: **Low** — The low respiration rate indicates less active respiration in this dataset, alongside the listed packaging recommendation.
- Ethylene production / sensitivity: **Very Low / Low** — Ethylene production is Very Low and ethylene sensitivity is Low in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **81%**
- pH level: **3.8**
- Chilling sensitivity: **No**
- Mechanical fragility: **High**
- Recommended packaging: **PP Perforated Clamshell**

Summary: Grapes is recorded with a low respiration rate, very low ethylene production, low ethylene sensitivity, and 81% moisture content. Its chilling sensitivity is no and its mechanical fragility is high. In the classifier records, this trait combination is paired with PP Perforated Clamshell. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.18 Papaya

- Respiration rate: **High** — The high respiration rate indicates active respiration in this dataset, and the recommended package is the one paired with this trait profile.
- Ethylene production / sensitivity: **High / High** — Ethylene production is High and ethylene sensitivity is High in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **88%**
- pH level: **5.5**
- Chilling sensitivity: **Yes**
- Mechanical fragility: **Medium**
- Recommended packaging: **Corrugated Board**

Summary: Papaya is recorded with a high respiration rate, high ethylene production, high ethylene sensitivity, and 88% moisture content. Its chilling sensitivity is yes and its mechanical fragility is medium. In the classifier records, this trait combination is paired with Corrugated Board. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.19 Green Peas

- Respiration rate: **Very High** — The very high respiration rate indicates a strongly active commodity in this dataset, so its packaging recommendation is paired with a trait profile that requires the listed packaging choice.
- Ethylene production / sensitivity: **Very Low / Medium** — Ethylene production is Very Low and ethylene sensitivity is Medium in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **78%**
- pH level: **6.3**
- Chilling sensitivity: **No**
- Mechanical fragility: **Low**
- Recommended packaging: **LDPE**

Summary: Green Peas is recorded with a very high respiration rate, very low ethylene production, medium ethylene sensitivity, and 78% moisture content. Its chilling sensitivity is no and its mechanical fragility is low. In the classifier records, this trait combination is paired with LDPE. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.20 Cauliflower

- Respiration rate: **Medium** — The medium respiration rate indicates an intermediate respiration level in this dataset, alongside the listed packaging recommendation.
- Ethylene production / sensitivity: **Low / High** — Ethylene production is Low and ethylene sensitivity is High in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **92%**
- pH level: **6.2**
- Chilling sensitivity: **No**
- Mechanical fragility: **Medium**
- Recommended packaging: **LDPE Wrap**

Summary: Cauliflower is recorded with a medium respiration rate, low ethylene production, high ethylene sensitivity, and 92% moisture content. Its chilling sensitivity is no and its mechanical fragility is medium. In the classifier records, this trait combination is paired with LDPE Wrap. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.21 Okra

- Respiration rate: **High** — The high respiration rate indicates active respiration in this dataset, and the recommended package is the one paired with this trait profile.
- Ethylene production / sensitivity: **Low / Medium** — Ethylene production is Low and ethylene sensitivity is Medium in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **90%**
- pH level: **6**
- Chilling sensitivity: **Yes**
- Mechanical fragility: **Medium**
- Recommended packaging: **PP Perforated**

Summary: Okra is recorded with a high respiration rate, low ethylene production, medium ethylene sensitivity, and 90% moisture content. Its chilling sensitivity is yes and its mechanical fragility is medium. In the classifier records, this trait combination is paired with PP Perforated. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.22 Brinjal

- Respiration rate: **Medium** — The medium respiration rate indicates an intermediate respiration level in this dataset, alongside the listed packaging recommendation.
- Ethylene production / sensitivity: **Low / High** — Ethylene production is Low and ethylene sensitivity is High in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **92%**
- pH level: **5.5**
- Chilling sensitivity: **Yes**
- Mechanical fragility: **Medium**
- Recommended packaging: **LDPE**

Summary: Brinjal is recorded with a medium respiration rate, low ethylene production, high ethylene sensitivity, and 92% moisture content. Its chilling sensitivity is yes and its mechanical fragility is medium. In the classifier records, this trait combination is paired with LDPE. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.23 Guava

- Respiration rate: **High** — The high respiration rate indicates active respiration in this dataset, and the recommended package is the one paired with this trait profile.
- Ethylene production / sensitivity: **Medium / Medium** — Ethylene production is Medium and ethylene sensitivity is Medium in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **83%**
- pH level: **4**
- Chilling sensitivity: **Yes**
- Mechanical fragility: **Medium**
- Recommended packaging: **LDPE Shrink Wrap**

Summary: Guava is recorded with a high respiration rate, medium ethylene production, medium ethylene sensitivity, and 83% moisture content. Its chilling sensitivity is yes and its mechanical fragility is medium. In the classifier records, this trait combination is paired with LDPE Shrink Wrap. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.24 Pomegranate

- Respiration rate: **Low** — The low respiration rate indicates less active respiration in this dataset, alongside the listed packaging recommendation.
- Ethylene production / sensitivity: **Low / Low** — Ethylene production is Low and ethylene sensitivity is Low in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **78%**
- pH level: **3**
- Chilling sensitivity: **No**
- Mechanical fragility: **Low**
- Recommended packaging: **Corrugated Board**

Summary: Pomegranate is recorded with a low respiration rate, low ethylene production, low ethylene sensitivity, and 78% moisture content. Its chilling sensitivity is no and its mechanical fragility is low. In the classifier records, this trait combination is paired with Corrugated Board. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

2.25 Green Chilli

- Respiration rate: **Medium** — The medium respiration rate indicates an intermediate respiration level in this dataset, alongside the listed packaging recommendation.
- Ethylene production / sensitivity: **Low / Medium** — Ethylene production is Low and ethylene sensitivity is Medium in the dataset; these two recorded traits are part of the profile used alongside the recommended packaging.
- Moisture content: **88%**
- pH level: **5.3**
- Chilling sensitivity: **Yes**
- Mechanical fragility: **Low**
- Recommended packaging: **LDPE**

Summary: Green Chilli is recorded with a medium respiration rate, low ethylene production, medium ethylene sensitivity, and 88% moisture content. Its chilling sensitivity is yes and its mechanical fragility is low. In the classifier records, this trait combination is paired with LDPE. The recommendation therefore follows the specific trait profile present in the dataset rather than a general rule about all foods.

3.1 PP Perforated Clamshell

- Material type: **PP Perforated Clamshell**
- OTR: **8000 cc/m²/day**
- WVTR: **500 g/m²/day**

Dataset fit: This material is paired with Strawberry, Grapes in the shelf-life records. Its observed OTR is 8000 cc/m²/day and its observed WVTR is 500 g/m²/day. Within the metric definitions used in this reference, lower OTR indicates a better oxygen barrier and lower WVTR indicates less water-vapor transmission.

3.2 MAP High Barrier

- Material type: **MAP High Barrier**
- OTR: **6–15 cc/m²/day**
- WVTR: **2–5 g/m²/day**

Dataset fit: This material is paired with Strawberry, Blueberry, Spinach, Tomato, Broccoli, Mango, Banana, Apple, Carrot, Cabbage, Lettuce, Cucumber, Bell Pepper, Grapes, Papaya, Green Peas, Cauliflower, Okra, Brinjal, Guava, Pomegranate, Green Chilli in the shelf-life records. Its observed OTR is 6–15 cc/m²/day and its observed WVTR is 2–5 g/m²/day. Within the metric definitions used in this reference, lower OTR indicates a better oxygen barrier and lower WVTR indicates less water-vapor transmission. The material has multiple observed barrier values in the supplied records, so the range is reported here.

3.3 LDPE Vented

- Material type: **LDPE Vented**
- OTR: **8000 cc/m²/day**
- WVTR: **480 g/m²/day**

Dataset fit: This material is paired with Blueberry in the shelf-life records. Its observed OTR is 8000 cc/m²/day and its observed WVTR is 480 g/m²/day. Within the metric definitions used in this reference, lower OTR indicates a better oxygen barrier and lower WVTR indicates less water-vapor transmission.

3.4 LDPE Perforated

- Material type: **LDPE Perforated**
- OTR: **7500–8500 cc/m²/day**
- WVTR: **18–520 g/m²/day**

Dataset fit: This material is paired with Spinach, Broccoli, Carrot, Cabbage, Lettuce in the shelf-life records. Its observed OTR is 7500–8500 cc/m²/day and its observed WVTR is 18–520 g/m²/day. Within the metric definitions used in this reference, lower OTR indicates a better oxygen barrier and lower WVTR indicates less water-vapor transmission. The material has multiple observed barrier values in the supplied records, so the range is reported here.

3.5 LDPE

- Material type: **LDPE**
- OTR: **7000–7500 cc/m²/day**
- WVTR: **15–18 g/m²/day**

Dataset fit: This material is paired with Tomato, Banana, Apple, Onion, Garlic, Bell Pepper, Green Peas, Brinjal, Green Chilli in the shelf-life records. Its observed OTR is 7000–7500 cc/m²/day and its observed WVTR is 15–18 g/m²/day. Within the metric definitions used in this reference, lower OTR indicates a better oxygen barrier and lower WVTR indicates less water-vapor transmission. The material has multiple observed barrier values in the supplied records, so the range is reported here.

3.6 Corrugated Board

- Material type: **Corrugated Board**
- OTR: **60000 cc/m²/day**
- WVTR: **15000 g/m²/day**

Dataset fit: This material is paired with Mango, Potato, Papaya, Pomegranate in the shelf-life records. Its observed OTR is 60000 cc/m²/day and its observed WVTR is 15000 g/m²/day. Within the metric definitions used in this reference, lower OTR indicates a better oxygen barrier and lower WVTR indicates less water-vapor transmission.

3.7 HDPE Perforated

- Material type: **HDPE Perforated**
- OTR: **2000 cc/m²/day**
- WVTR: **10 g/m²/day**

Dataset fit: This material is paired with Potato in the shelf-life records. Its observed OTR is 2000 cc/m²/day and its observed WVTR is 10 g/m²/day. Within the metric definitions used in this reference, lower OTR indicates a better oxygen barrier and lower WVTR indicates less water-vapor transmission.

3.8 Mesh Bag

- Material type: **Mesh Bag**
- OTR: **60000 cc/m²/day**
- WVTR: **15000 g/m²/day**

Dataset fit: This material is paired with Onion, Garlic in the shelf-life records. Its observed OTR is 60000 cc/m²/day and its observed WVTR is 15000 g/m²/day. Within the metric definitions used in this reference, lower OTR indicates a better oxygen barrier and lower WVTR indicates less water-vapor transmission.

3.9 LDPE Shrink Wrap

- Material type: **LDPE Shrink Wrap**
- OTR: **7000 cc/m²/day**
- WVTR: **15 g/m²/day**

Dataset fit: This material is paired with Cucumber, Guava in the shelf-life records. Its observed OTR is 7000 cc/m²/day and its observed WVTR is 15 g/m²/day. Within the metric definitions used in this reference, lower OTR indicates a better oxygen barrier and lower WVTR indicates less water-vapor transmission.

3.10 LDPE Wrap

- Material type: **LDPE Wrap**
- OTR: **7000 cc/m²/day**
- WVTR: **18 g/m²/day**

Dataset fit: This material is paired with Cauliflower in the shelf-life records. Its observed OTR is 7000 cc/m²/day and its observed WVTR is 18 g/m²/day. Within the metric definitions used in this reference, lower OTR indicates a better oxygen barrier and lower WVTR indicates less water-vapor transmission.

3.11 PP Perforated

- Material type: **PP Perforated**
- OTR: **8000 cc/m²/day**
- WVTR: **500 g/m²/day**

Dataset fit: This material is paired with Okra in the shelf-life records. Its observed OTR is 8000 cc/m²/day and its observed WVTR is 500 g/m²/day. Within the metric definitions used in this reference, lower OTR indicates a better oxygen barrier and lower WVTR indicates less water-vapor transmission.

4. Storage Condition & Shelf-Life Patterns

The following findings describe observations in the 50 supplied shelf-life records. They do not treat differences between rows as causal unless the relevant conditions are held constant.

Longest observed shelf-life records

- Apple + MAP High Barrier: **180 days** at 2°C and 92% RH.
- Garlic + LDPE: **180 days** at 0°C and 65% RH.
- Garlic + Mesh Bag: **150 days** at 0°C and 65% RH.
- Onion + LDPE: **150 days** at 2°C and 70% RH.
- Apple + LDPE: **120 days** at 2°C and 92% RH.
- Onion + Mesh Bag: **120 days** at 2°C and 70% RH.
- Potato + HDPE Perforated: **120 days** at 8°C and 92% RH.
- Cabbage + MAP High Barrier: **90 days** at 1°C and 97% RH.

Shortest observed shelf-life records

- Spinach + LDPE Perforated: **6 days** at 2°C and 98% RH.
- Strawberry + PP Perforated Clamshell: **6 days** at 2°C and 90% RH.
- Okra + PP Perforated: **7 days** at 11°C and 92% RH.
- Blueberry + LDPE Vented: **10 days** at 1°C and 92% RH.
- Broccoli + LDPE Perforated: **10 days** at 1°C and 98% RH.
- Green Peas + LDPE: **10 days** at 1°C and 92% RH.
- Guava + LDPE Shrink Wrap: **10 days** at 9°C and 90% RH.
- Lettuce + LDPE Perforated: **10 days** at 1°C and 97% RH.

Representative commodity comparisons

- Strawberry: the observed records pair PP Perforated Clamshell with 2°C/90% RH and 6 days, while MAP High Barrier is paired with 2°C/95% RH and 12 days.
- Apple: under the same 2°C and 92% RH conditions, LDPE produced 120 days versus 180 days for MAP High Barrier.
- Tomato: the observed records pair LDPE with 13°C/90% RH and 12 days, while MAP High Barrier is paired with 8°C/90% RH and 20 days.
- Onion: under the same 2°C and 70% RH conditions, Mesh Bag produced 120 days versus 150 days for LDPE.
- Banana: under the same 14°C and 90% RH conditions, LDPE produced 14 days versus 24 days for MAP High Barrier.

What the records show about temperature and humidity

Across all 50 rows, shelf life has a descriptive correlation of -0.33 with storage temperature and -0.60 with relative humidity. These are dataset-level associations only: food type, packaging material, and storage conditions change together across many rows, so the correlations should not be read as isolated causal effects.

5. Selection Guidelines

These guidelines are deliberately limited to relationships visible in the classifier records. They are not general packaging-engineering rules beyond what the supplied data can support.

If respiration is Very High

The dataset includes Strawberry, Spinach, Broccoli, and Green Peas. Their recorded recommendations are PP Perforated Clamshell / MAP High Barrier, LDPE Perforated / MAP High Barrier, LDPE Perforated / MAP High Barrier, and LDPE / MAP High Barrier respectively. The data therefore shows multiple package choices for the same respiration category.

If ethylene production is High

Banana, Mango, Papaya, and Tomato have High ethylene production. Their recorded packages include LDPE and MAP High Barrier for Banana; Corrugated Board and MAP High Barrier for Mango; Corrugated Board and MAP High Barrier for Papaya; and LDPE and MAP High Barrier for Tomato.

If ethylene sensitivity is High

Broccoli, Spinach, Lettuce, Mango, Banana, Papaya, Brinjal, Cucumber, Cauliflower, and other listed commodities carry High sensitivity. The associated recommendations vary, so ethylene sensitivity should be used together with the other recorded traits rather than as a standalone package selector.

If mechanical fragility is High

Strawberry, Blueberry, and Grapes are the High-fragility commodities in the classifier records. Their listed packaging includes PP Perforated Clamshell, MAP High Barrier, LDPE Vented, and PP Perforated.

If moisture content is high

The classifier contains commodities from 88% through 96% moisture, among others. The dataset pairs these foods with several package types, including perforated/vented LDPE, MAP High Barrier, LDPE Wrap, and PP-based packaging; moisture alone does not identify one unique material in the supplied records.

If chilling sensitivity is Yes

Banana, Tomato, Mango, Potato, Cucumber, Bell Pepper, Papaya, Okra, Brinjal, Guava, and Green Chilli are marked Yes. Their recorded packaging recommendations vary, so chilling sensitivity should be considered as one profile field alongside respiration, ethylene, moisture, and fragility.

Barrier-property guide from the shelf-life dataset

MAP High Barrier has the lowest observed OTR range (6–15 cc/m²/day) and lowest observed WVTR range (2–5 g/m²/day) among the listed materials. Corrugated Board and Mesh Bag each have 60,000 cc/m²/day OTR and 15,000 g/m²/day WVTR in the records. The dataset therefore provides a clear numerical contrast in barrier values, while shelf-life outcomes still depend on the particular food, conditions, and packaging combination.

6. Frequently Asked Questions

1. What packaging is recorded for Strawberry?

The classifier dataset pairs Strawberry with PP Perforated Clamshell and MAP High Barrier. Its profile is very high respiration, low ethylene production, low ethylene sensitivity, 91% moisture, no chilling sensitivity, and high mechanical fragility.

2. What packaging is recorded for Banana?

Banana is paired with LDPE and MAP High Barrier in the classifier records. The profile shows high respiration, high ethylene production, high ethylene sensitivity, 75% moisture, chilling sensitivity marked Yes, and medium mechanical fragility.

3. Why does ethylene sensitivity matter in this reference?

Ethylene sensitivity is one of the recorded classifier traits and varies by commodity. For example, Banana, Mango and Papaya are marked High, while Strawberry and Blueberry are marked Low; the packaging recommendations are paired with those full profiles.

4. Which material has the lowest observed OTR?

MAP High Barrier has observed OTR values from 6 to 15 cc/m²/day, the lowest range recorded among the packaging materials in the shelf-life dataset. In the dataset, lower OTR values correspond to stronger oxygen-barrier performance as an interpretation of the OTR number itself.

5. Which material has the lowest observed WVTR?

MAP High Barrier has observed WVTR values from 2 to 5 g/m²/day, the lowest range in the dataset. A lower WVTR means less water-vapor transmission according to the meaning of the metric.

6. What packaging is recorded for high-respiration foods?

The dataset contains several high or very high respiration foods with different recommendations. Strawberry uses PP Perforated Clamshell and MAP High Barrier; Spinach uses LDPE Perforated and MAP High Barrier; Green Peas uses LDPE and MAP High Barrier, showing that respiration rate alone does not determine one package.

6. Frequently Asked Questions

7. What is the longest observed shelf life?

The maximum observed shelf life is 180 days. It occurs for Apple with MAP High Barrier at 2°C and 92% relative humidity, and for Garlic with LDPE at 0°C and 65% relative humidity.

8. What happened for Apple?

Apple has two records at 2°C and 92% RH: LDPE gives 120 days and MAP High Barrier gives 180 days. The difference is observed under the same recorded storage conditions, with packaging material changing.

9. What happened for Tomato?

Tomato has an LDPE record at 13°C and 90% RH with 12 days, and a MAP High Barrier record at 8°C and 90% RH with 20 days. Because both packaging and temperature differ, these records do not isolate the effect of temperature alone.

10. Does higher humidity always mean longer shelf life?

The supplied records do not support a universal rule that higher humidity always increases shelf life. For example, the dataset contains different foods and packaging at different humidity levels, so humidity effects cannot be separated from food and packaging differences across the full table.

11. What does OTR mean here?

OTR is the Oxygen Transmission Rate recorded in cc/m²/day. The material records show values such as 6–15 for MAP High Barrier and 60,000 for Corrugated Board and Mesh Bag; within this reference, a lower OTR is interpreted as a better oxygen barrier.

12. What does WVTR mean here?

WVTR is the Water Vapor Transmission Rate recorded in g/m²/day. The dataset shows 2–5 for MAP High Barrier and 15,000 for Corrugated Board and Mesh Bag; within this reference, lower WVTR is interpreted as less water-vapor transmission.

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